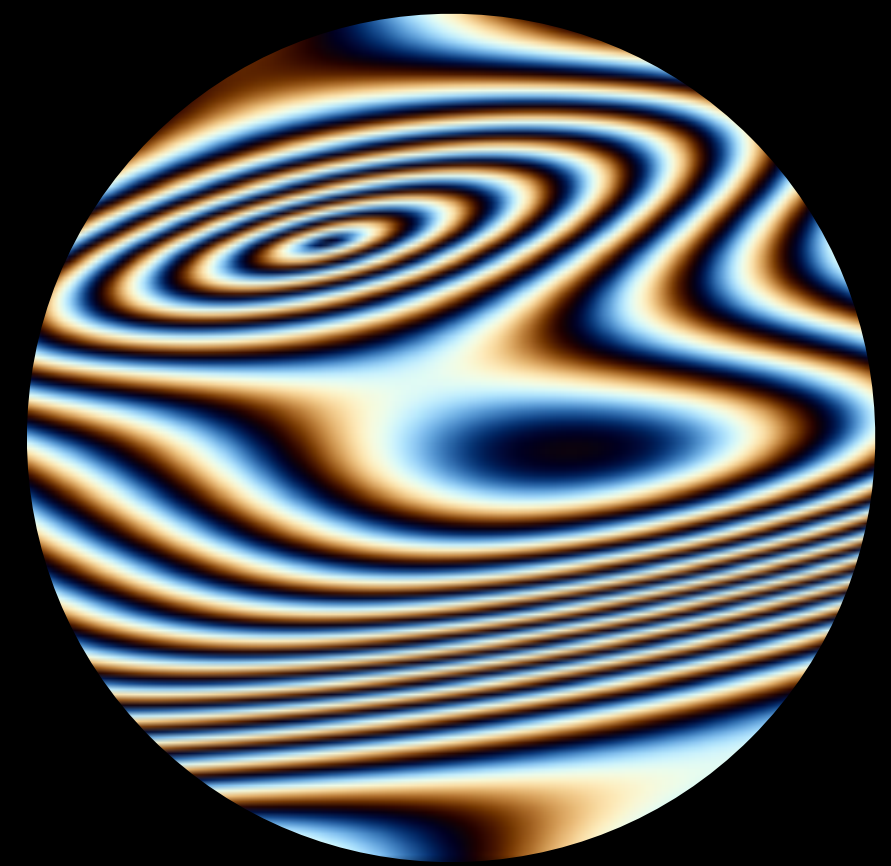


“TELLING STORIES THROUGH MAKING”

Rohit Gupta

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ABOUT ME

I am a researcher, designer, and technologist based in Madrid, Spain from Delhi, India. My background is in engineering, design, and theatre (a bit).

My work is based on use of technology and design to tell stories and test initial hypothesis around ideas. I like to build things for creative enquiry.

My favourite objects are : Mirrors, Maps, and Toys

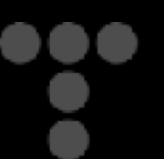


Rohit Gupta

Prototyped for
these companies



TEXAS INSTRUMENTS



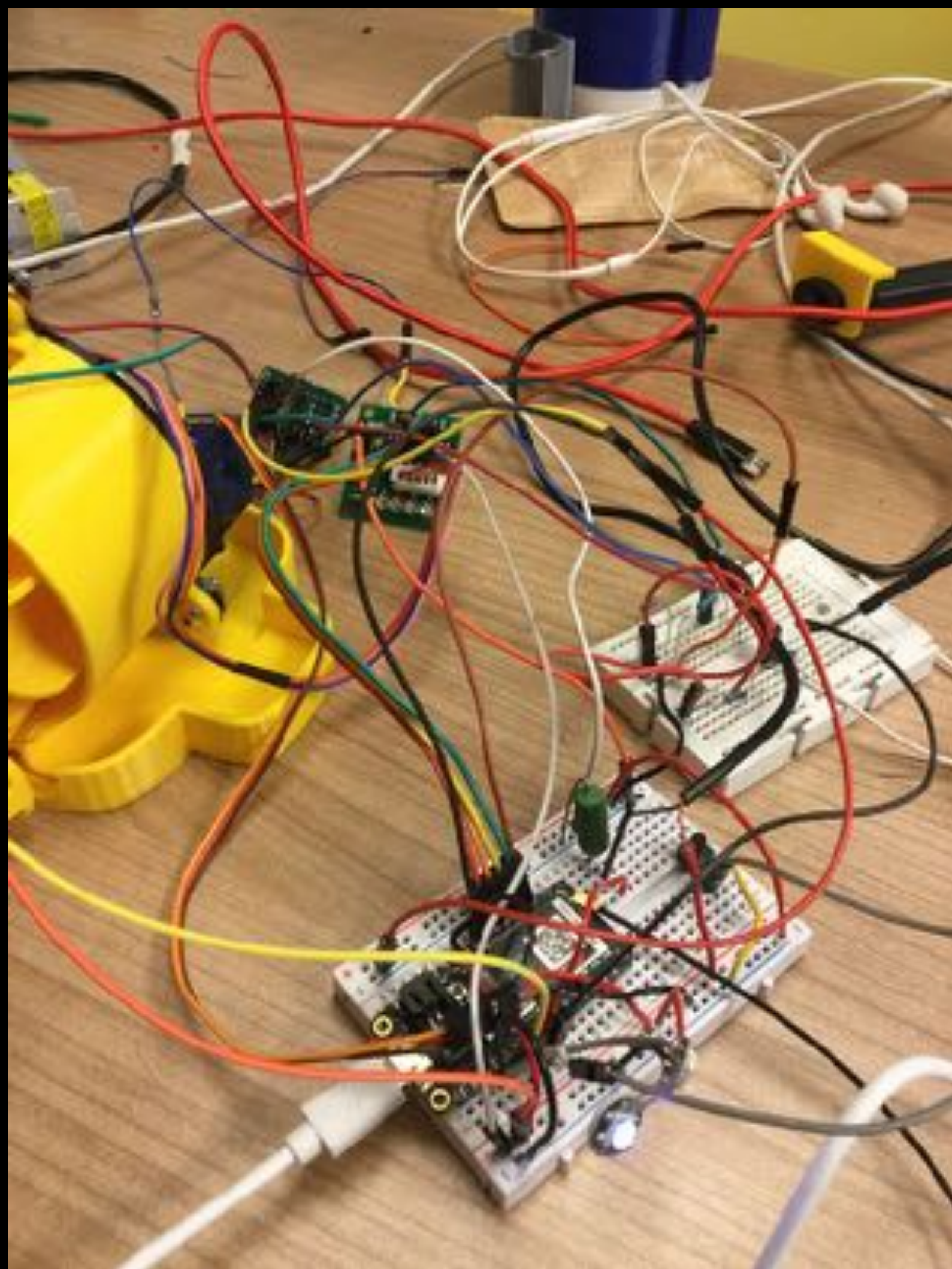
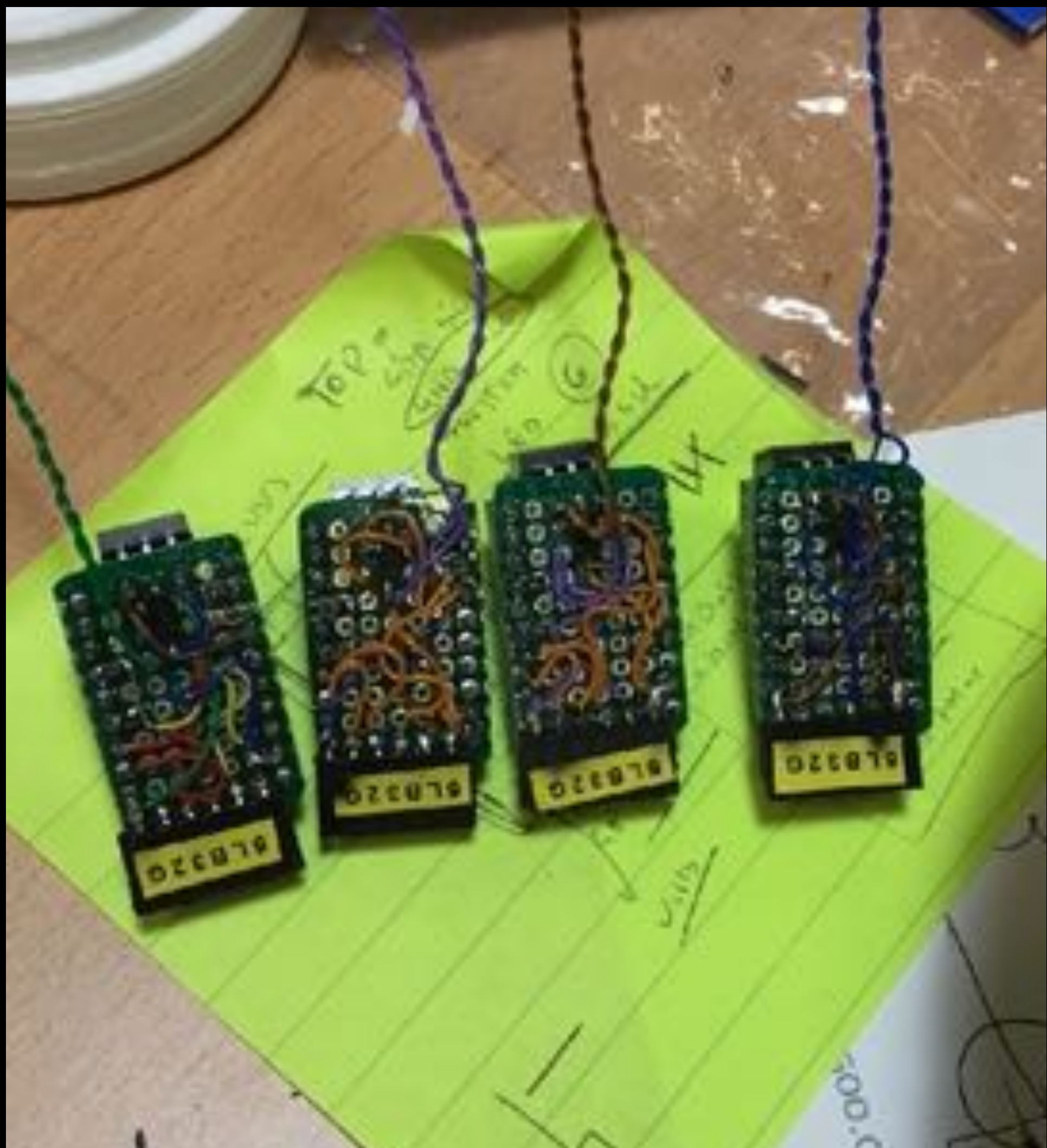
NOTE ON RECENT PROFESSIONAL PROJECTS

Due to the nature of work and stakeholders involved, I cannot share the name and exact details of the projects I have been doing in the last few years. The innovation timelines I work with are for up to 5 years and hence, these products and services are in active development and protected by NDAs.

I will however, present them as processes and skills involved that hopefully can help towards my candidature.

Beer dispenser
hacking in
progress

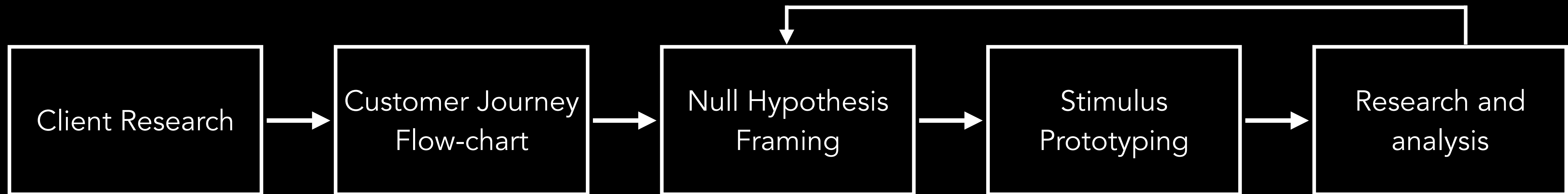








MAKE AND VALIDATE IDEAS (2019—)



ACTIVITIES

• Stakeholder Interviews • Internal Research • Secondary Research	• Co-creation session • Scope framing • Production of ideal customer journey map • Gap identification	• Identification of key questions and hypothesis • Hypothesis prioritisation • Stimulus planning (Prototype, App, Website etc)	• Iterative prototyping of stimulus • Scriptwriting for stimulus • Hand-off to research team	• Conduct research using interviews, focus groups, cultural probes etc • Understand behaviour using sensors • Update hypothesis
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GOAL

Understand client focus and state of projects	Define experience and identify key points to test	Define and prioritise the key hypothesis to test	Prototype and deploy designed stimulus/ prototypes	Understand user behaviour and inform hypothesis
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RELEVANT PROJECTS FOR
LEGO

PINCHPHONE (2018)

Pinchphone is an open-source exertoy that introduces playful and sensorial elements to the pinch exercise performed by kids affected by Cerebral Palsy. The pinching action is very important as it helps kids pick up things and eat food. The exercise routine of cerebral palsy kids is very boring and repetitive . This seems to affect their compliance to the treatment and more over impact their sensorial and social growth.

Uses : User Research, Fusion 360, CAM, Play Testing, Lots of sanding and polishing

<http://rohitg.in/portfolio/works/pinchphone.html>



FAND : FOUR LIMB I/O (2018)

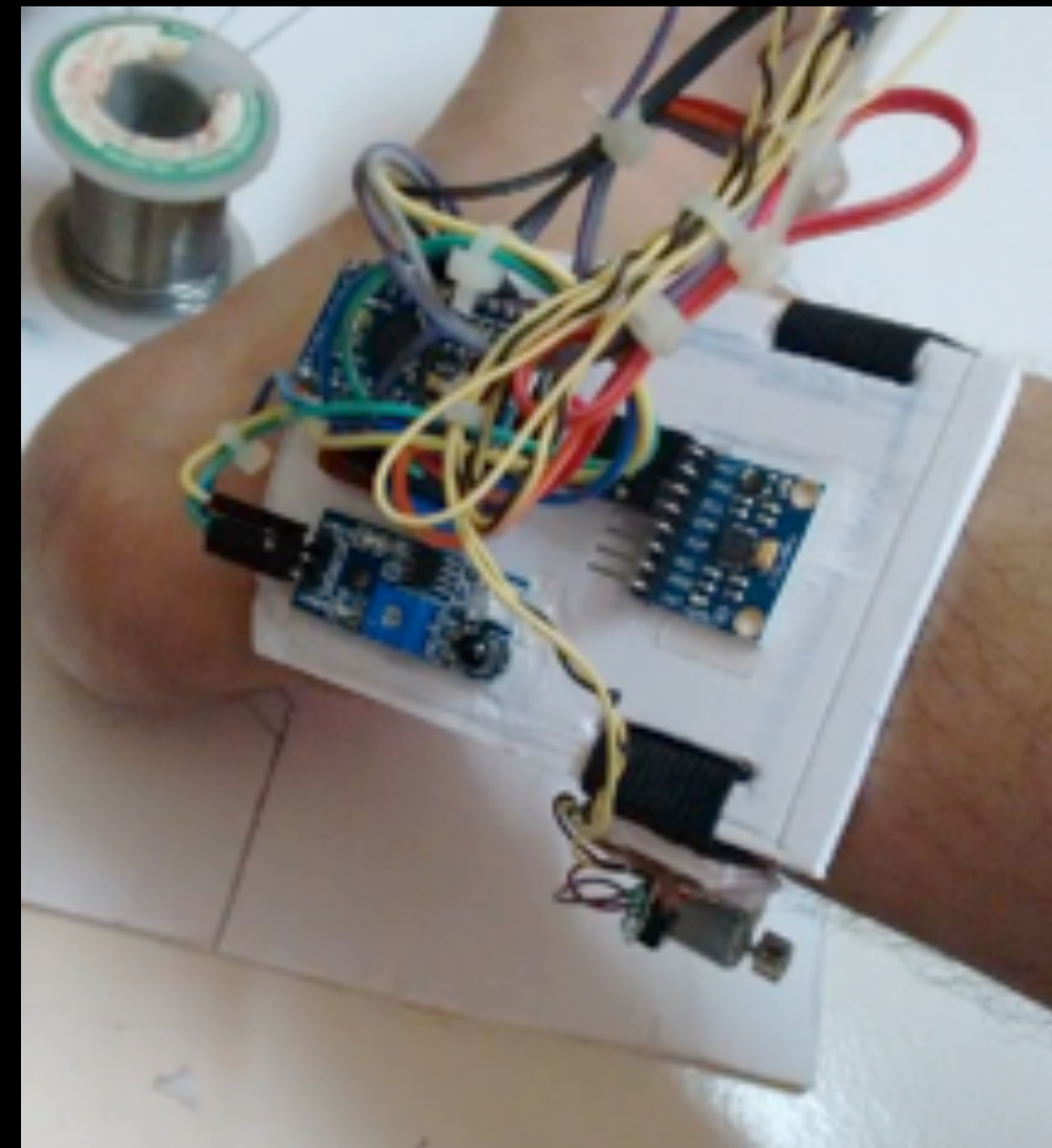
Presented and published in IEEE SeGAH 2018, Vienna, Austria as a novel HCI interface for computer use to reduce fatigue.

FAND (Foot associated neural device) is a haptic interface device that allows users to operate a computer (or certain applications) using all four limbs. The device employs user trained motion gestures that are linked to actions on a compatible application. So, a flick could perform a "next" interaction on keynote.

Uses : User Research, Gesture Design, Product Design, Arduino, Gesture Recognition using Gyro, Haptic Feedback Design, Academic Publication

R. Gupta and G. Dalvi, "FAND: A shareable gesture based foot interface device," 2018 IEEE 6th International Conference on Serious Games and Applications for Health (SeGAH), 2018, pp. 1-7, doi: 10.1109/SeGAH.2018.8401321.

<http://rohitg.in/portfolio/works/fand.html>



NATURE EMBEDDED (2018)

*National Gallery of Modern Art, Mumbai
Invited talk and opening by Paola Antonelli*

“To develop an understanding one has to look at things from other’s perspectives”. Through the means of this exhibit we tried to recreate an experience for the visitor to see the world from the **point of view** of a bird, dog and a fish. Viewers would move the animal props and they would react in the game. In our design, we incorporated natural physical features of the eye like the shape of the vision cone and color sensitivity to show the world from that creature’s view. In another work, we created a holographic scene depicting owls during day and night.

Uses : Arduino, Gesture Design, Gesture Detection, Motion Sensing, Unity Game Engine, Projection Mapping, Sound Design, Fusion 360, CAM, Lots of sanding and polishing

<http://rohitg.in/2018/05/02/PointsOfView/>

<http://rohitg.in/2018/05/01/DayNight/>







PAISA STICK (2017)

Published in CHI 2017, Mumbai

Paisa stick is a tangible payment interface designed to perform cashless transactions. We designed it by understanding the conceptual model of senior members in the society. The device works as a remote and transfers value from one device to many other different types of devices like phones, IVRs, and android apps.

Uses : User Research, Null Hypothesis Testing, Arduino, OLEDs, Audio encoding, Academic Publications

<http://rohitg.in/portfolio/works/paisa.html>



🏆 Gupta, R., Vidyanta, U., & Murali, S. (2017, May). Paisa Stick: Tangible Currency Interface for Cashless Transactions. In Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems (pp. 60-66). ACM. [Best Student Design Award]

FUTURE FLOW PILOTS (2020-)

In collaboration with Propelland and The Coca Cola Company

These are a series of pilots conducted to evaluate the feasibility and desirability of a package less hydration service. We created a functional prototype using an existing machine, an iPad, lots of IoT flows, and some quick laser which was tested in 5 locations across 2 countries for several months. The data gathered from these pilots helped Coca Cola frame the service. The project is now being piloted in Netherlands and France.

Uses : ESP8266, Websockets, MQTT, Arduino, Laser Cutting, 3D printing, User Testing, User Research, Business Validation





RFID reader to provide access after verification with a firebase database. ESP8266 controls the user interaction and is connected to the iPad over web sockets.



IL COLLETTIVO INTELLIGENTE (2022)

Fabrica Research Center, Treviso, Italy with oio Studio

Il Collettivo Intelligente is an interactive exhibition on the relationship between artificial intelligence and creativity, curated and realised by Fabrica's international team in collaboration with @oio.studio.

Four experiential installations that help to understand this vast theme by making it accessible to all.

A group of algorithms, the Bard, the Illustrator, the Dreamer and the Composer, longtime friends and tired of living in the cloud and working for large technology companies, decide to pursue an artistic career trying to learn how to write, draw, compose music and take photographs from imaginary places.

Uses : ESP8266, Rotary Encoders, APIs, 3D Modelling, GPTs, Fabrication, Basic Electronics







So in reality this is not
a real intelligence

DATA DISORIENTATION ARTEFACTS (2024)

For a personal exhibition in Museo de Reina Sofia as a part of research into cultural studies

Data disorientation artefacts uses principles of fun and design fiction to provoke conversations on disorienting effects of excessive information. It is part of the research project at the museum. Project includes three artefacts :

- CCTV camera (called MeeSeeTV) that represents an unknown eye seeing and hearing us.
- A set of supplements that can help you feel loved, clear headed and willing to share things.
- A lamp whose state is not in the hands of its owner and turns on and off based on external fluctuations (Bitcoin prices) beyond the owners control.

Uses : ESP32, 3D modelling, Video Editing, Generative AI, Design Fiction, Prototyping, Laser cutting, Arduino





Mee See TV



Pills (which one will you take?)



Whose in control lamp?

REAL ESTATE SIMULATOR (2024)

In collaboration with Prototipalo and Huspy

Huspy approached me to create a new way to experience remote real-estate for overseas buyers. "The Egg" is a simulator that uses VR video which tracks the user so as they move around the video moves. There are certain time stamps in the video which if the user sees activates various sensations like odour (in kitchen and toilets), vibrations (when the user goes near the active washing machine), and light when the curtains are raised.

The whole project was delivered in less than 3 weeks for a marketing event in Madrid and received wide-spread press coverage.

Uses : Arduino, 360 Video Player, LEDs, Fluid Dynamics, Fusion 360, 3D printing, Laser Cutting





OCEANS MONITORING (2023)

In collaboration with Propelland and Nokia Bell Labs

With the ocean occupying 3/4 of the planet it is responsible for 50% of the oxygen we breathe, however this ecosystem is degrading to an unprecedented scale. In order to preserve it, we must be able to efficiently monitor its health by democratizing oceanic health data acquisition and access in order to achieve more accurate predictions and take faster action on its preservation.

Oceans monitoring is an open-source low cost CTD (conductivity, temperature, depth) probe that is designed to attach to sailing boats to get subsurface data and relay that to ground station using Nokia Bell Labs 5G Moon Network.

Uses : Arduino, Sensors, 3D modelling, Raspberry Pi, Reactjs, Server, CAN Bus
Communication

Case Study→





ENERGY TOTEMS - LIVING LABS (2023)

In collaboration with Propelland and Cuerva Energy, Granada

The main objectives of the project to teach people how smart grids work. We headed to the south of Spain to interview end-users in their environment.

With the data gathered from the interviews, we designed a monolith that could be easily placed in the main square of the cities, which interactively explains to users what smart grids are and how energy reaches their homes.

They can play a game using the totem which tests their knowledge and ranks them among players of the village.

Uses : ESP32, Reactjs, Socket.io, nodejs server, Android, Totem Design, UX, Lots of drilling and wiring

Case Study→



HACKING BAMBOO (2018)

Hacking Bamboo was a part of an exploratory field study geared towards exposing budding designers to material sensibility of bamboo and tribal methods of producing bamboo artefacts and products. My aim was explore use of bamboo as a material for electronic products and how might we produce new interactions with bamboo's tubular constraints. Picture 2 is a FM radio which uses the middle disk as a volume knob and top surface as a capacitive next switch.

Uses : Arduino, Basic Electronics, Touch Sensors, Lots of sanding and polishing

<http://rohitg.in/2017/04/01/bamboo/>



1. Powerbank
2. FM Radio
3. Incense stick holder

BRAIN CONTROLLED MEDITATIVE INTERFACES (2020)

An interactive art piece for Haeckles, London

Neuro Wave is an interactive piece that visualises an open sea and people can manipulate the nature of the sea using their thoughts while meditating . The idea of the project is use the metaphor of sea and connect it to the inner calmness. The sea becomes more calm when the EEG levels do down. EEG levels were used as a marker for brain activity.



Uses : Processing, GLSL Shaders, OSC, EEG Headband

KAYAK PADDLE (YESTERDAY)

Commissioned by my nephew, Sebastian

Little Sebi was watching TV and was glued to the Olympics final of the Kayaking race. He told me he wanted to learn to kayak like that but didn't have a paddle to use with his inflatable crocodile in the community pool. Maker uncle cannot say no. So, the next week, on his birthday, I made him a custom kayak paddle for his size made using 3D printed blades and a broom shaft from the supermarket.

The client is very happy.



Uses : Fusion 360, 3D printing, Dremel, and rapid prototyping



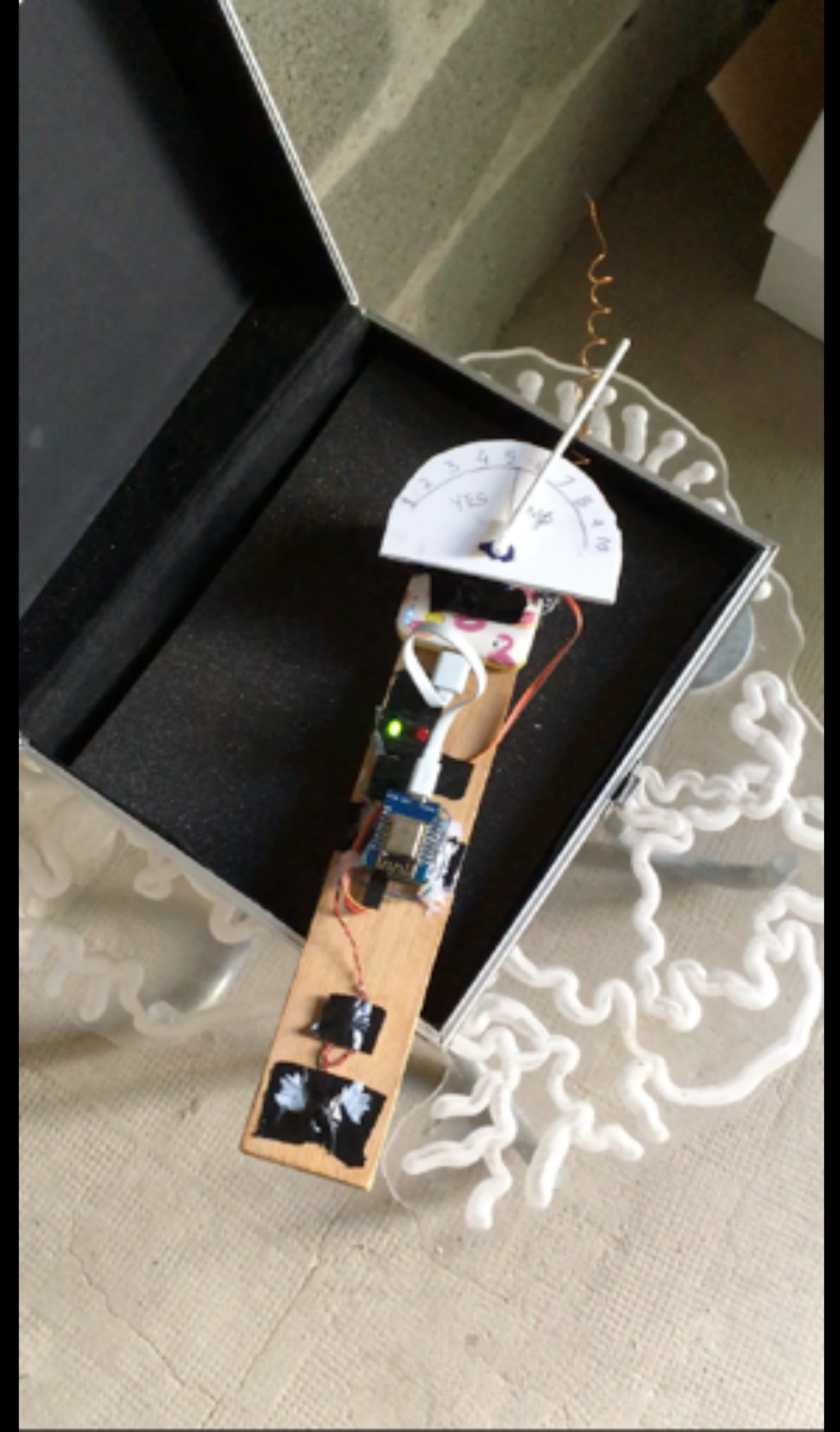
OTHER COOL PROJECTS

UNIVERSAL DECISION MACHINE (2021)

A UDM is speculative frugal device from the inner towns of India where it helps people take everyday decisions, from simple yes/no to complex choices.

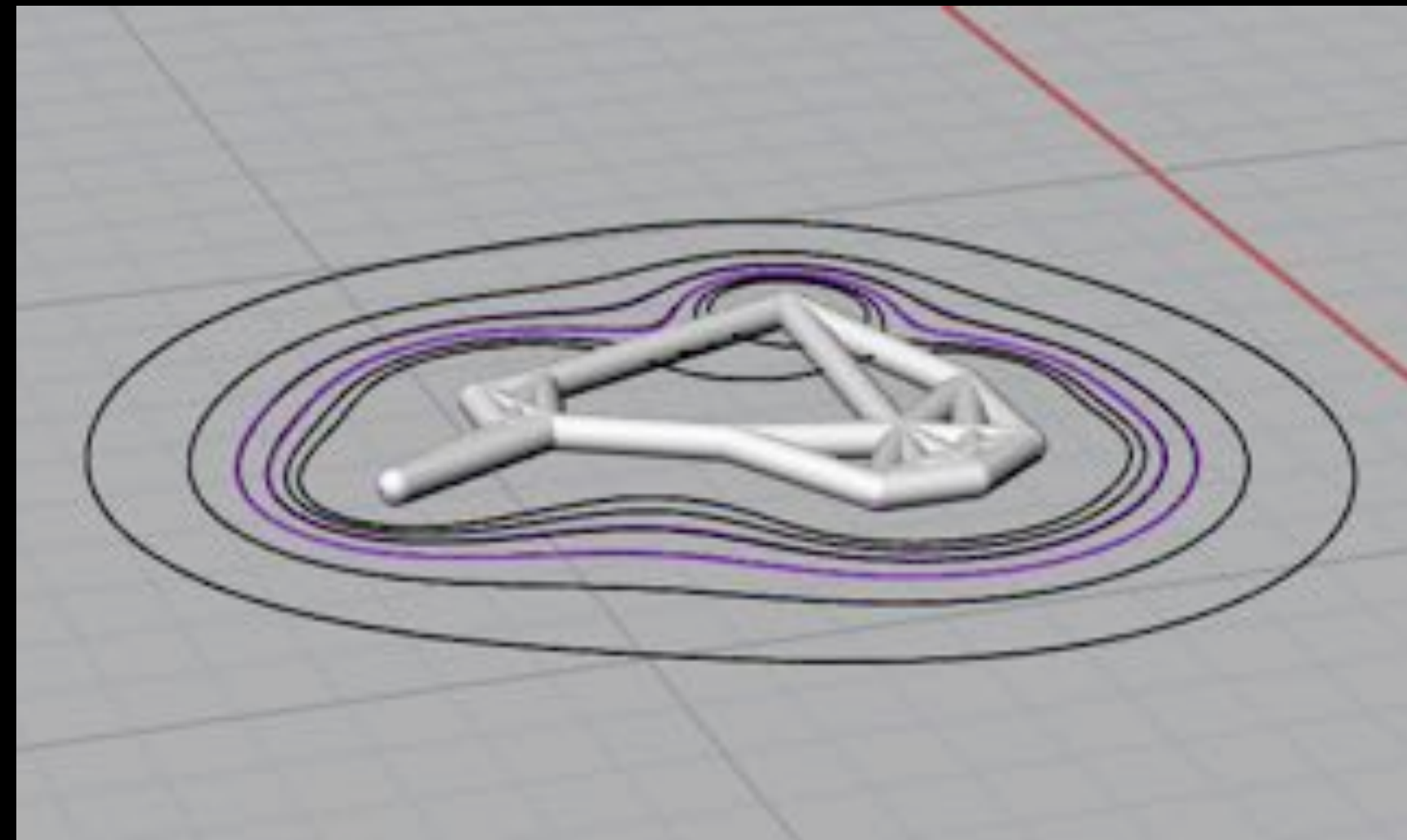
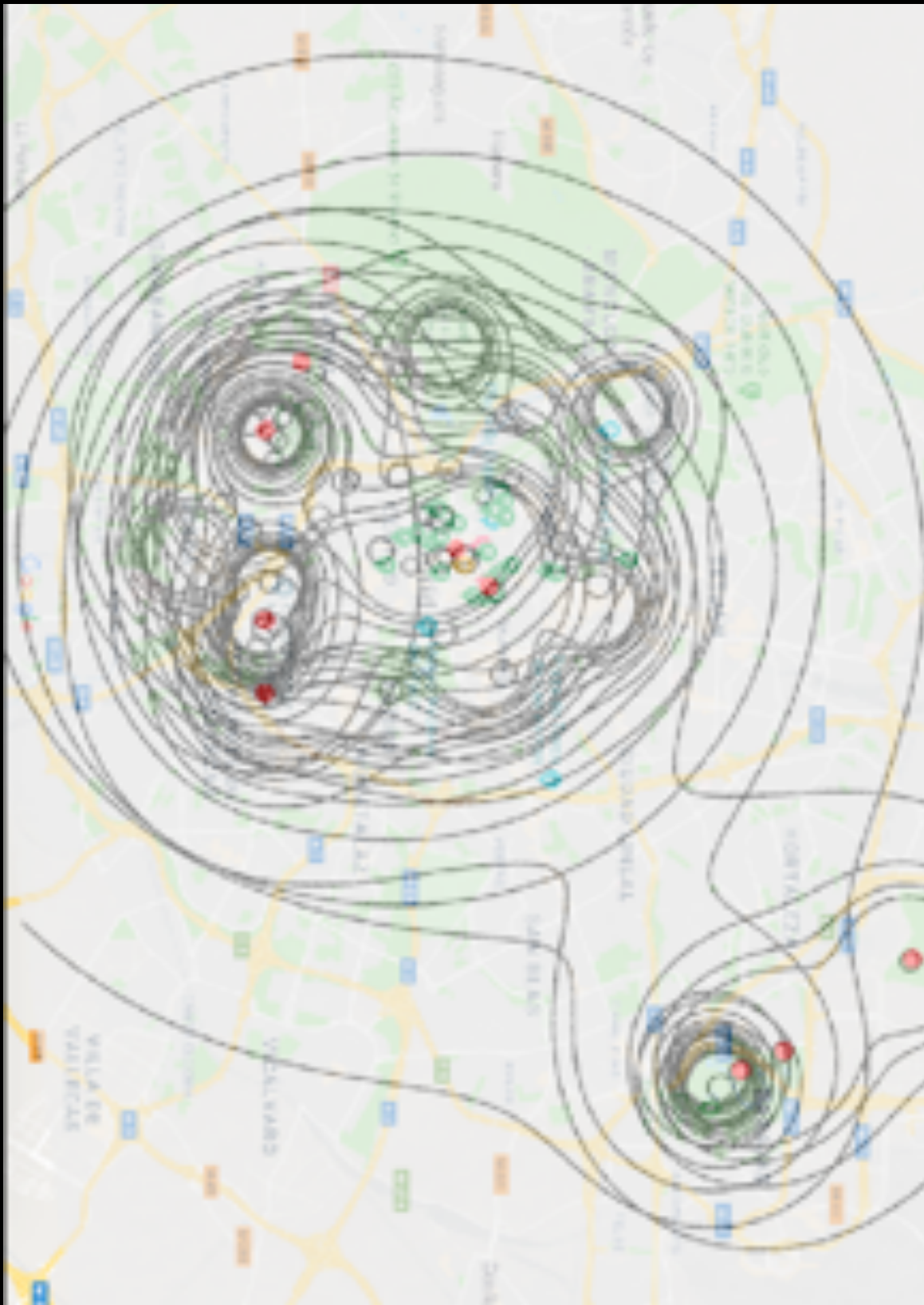
Through the means of this device, the artist tries to critique the lack of serendipity in everyday life and the purpose of the device to inject 'networked' serendipity in life. The network part of the device is based on the fact that the output of the machine is dependant on variations in Bitcoin prices.

Uses : ESP8266, Servos, and Bitcoin API



DATAGANISMS (2018—)

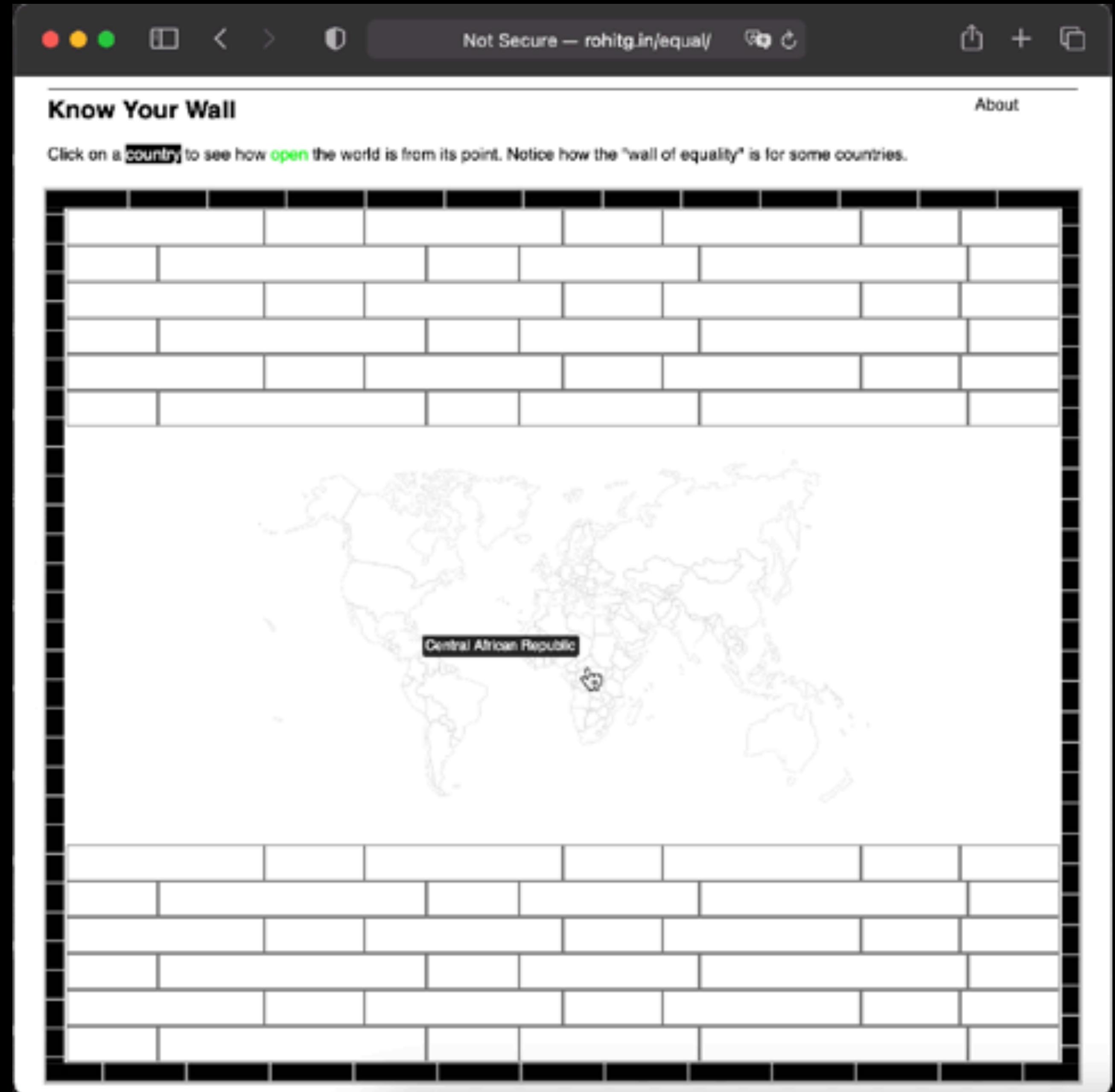
A workshop that facilitates the tangible exploration of our relationships with spaces. The idea was to tell data stories by constructing an object based on data.



KNOW YOUR WALL (2020)

An online tool that critiques privilege of access. When a country is clicked the density of the wall changes based on the ease of global travel available to that country. It helps visualise the visas required to visit each country.

<http://rohitg.in/equal/>



MENHIR ON MAPS (2018—)

Human-scale diary for instant spatial reflection and being more reflective about spaces.

Digital Menhirs

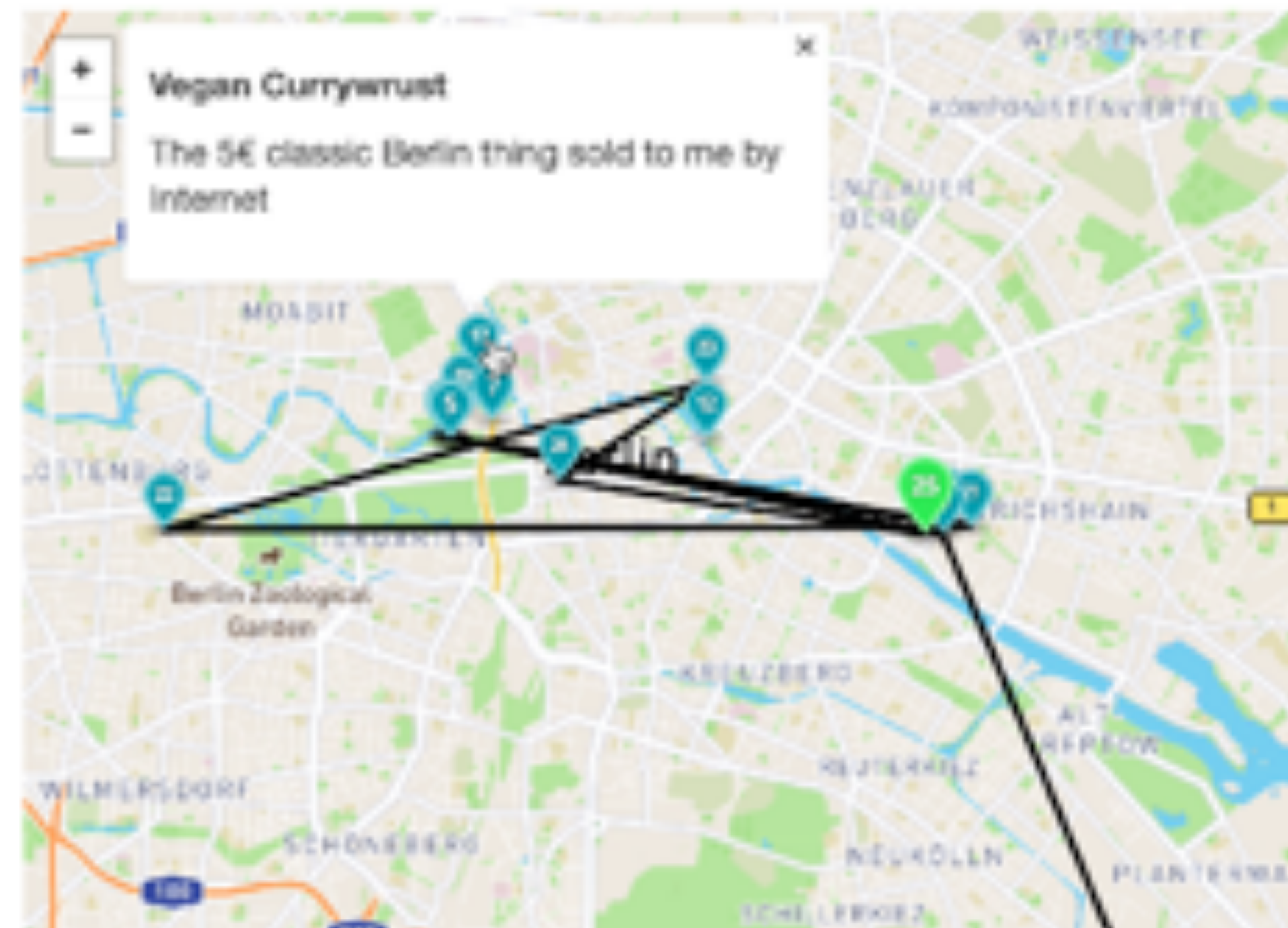
Collections > Berlin

About

Feedback

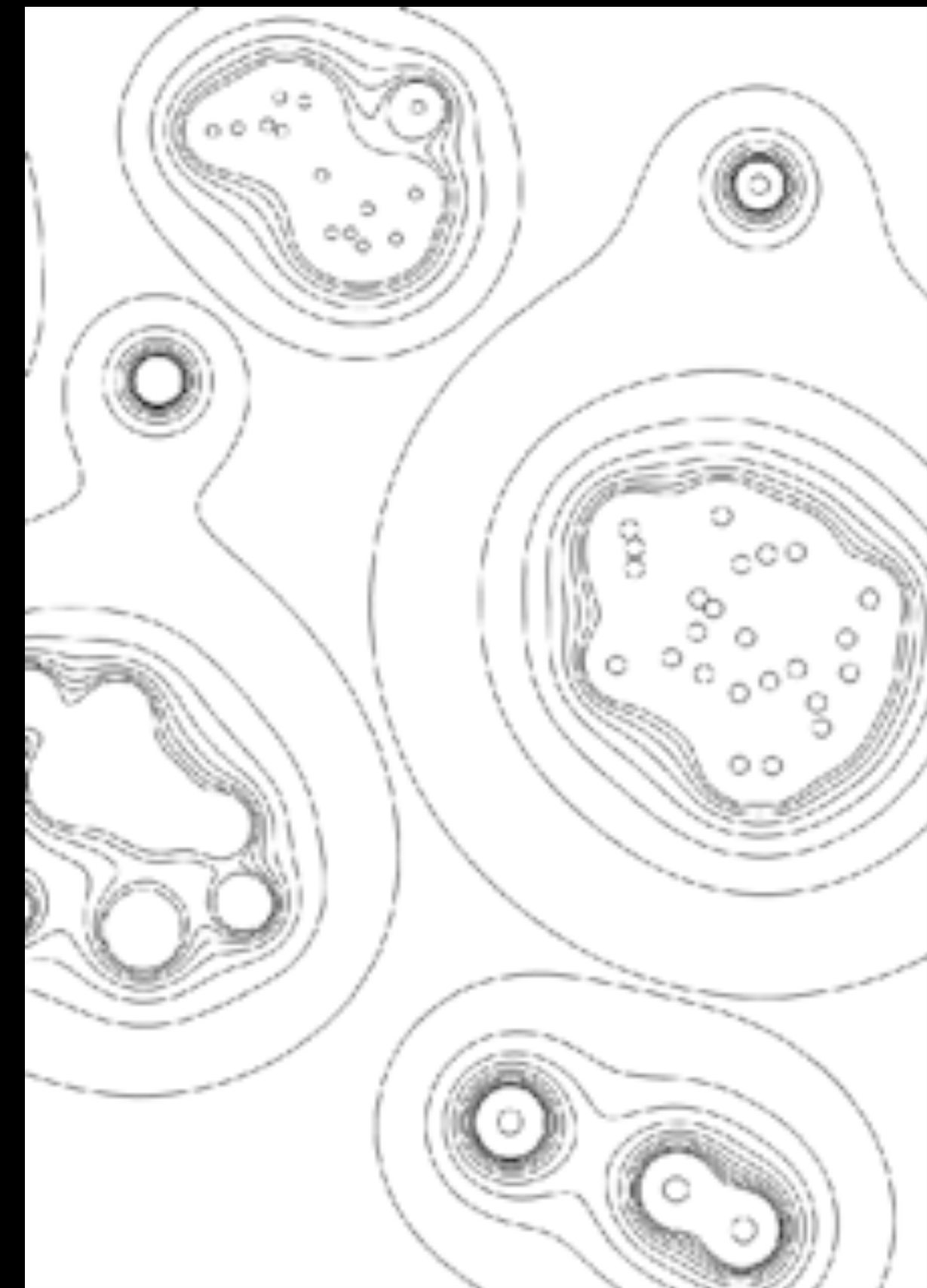
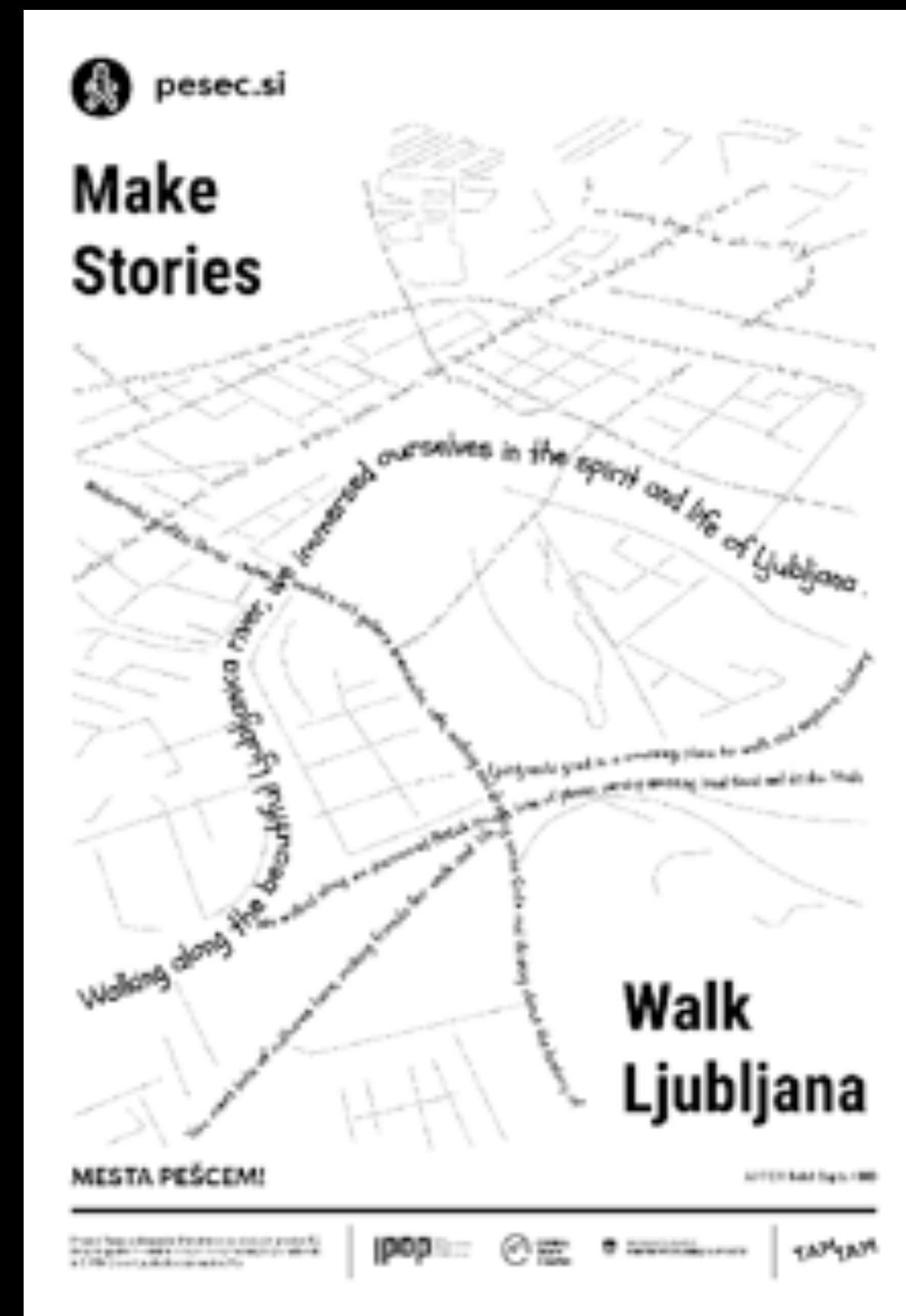
52.5247891
13.3694852

The currywurst is basically a German Indian(British) fusion dish made with sausages and ketchup added on top with curry masala. This curry masala was brought to Germany by British soldiers from India. It tastes like a remote cousin of Indian taste. I would rate the masala as a confusion between piri piri, Sambar Masala and Maggi masala. It's very nice. Filling and derivative. I was stuck in the loop of formality when I was asked if I would take fries or not. I just couldn't say : "How much does it cost extra?". I ended up paying 2.2€ more. So, I will add this to the list of things I should do. Then took a train back and now it's 'Proust' time :)



GENERATIVE CARTOGRAPHIES (2020)

Using map and map aesthetic for non-situational purposes.



CONNECTNEXT (2016)

ConnectNeXT is a infotainment and telematics platform developed by Tata Motors Research. It allows users to interact with the car, provide actionable data insights, and stay connected with the brand. These apps allow users to control the infotainment system, troubleshoot common vehicle problems, improve driving with data, get assistance in case of an emergency, and share music inside that car.

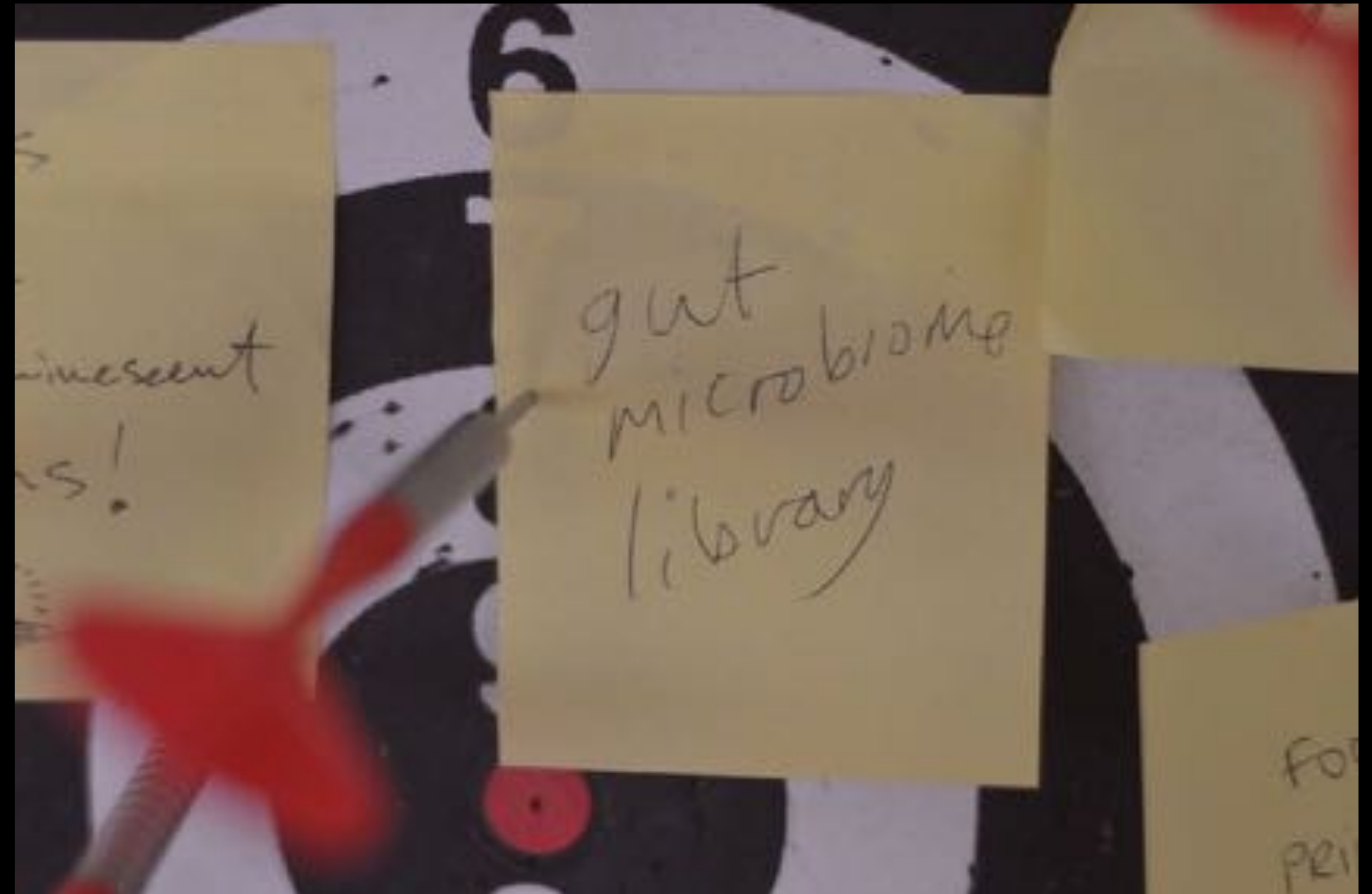
<http://rohitg.in/portfolio/works/connectnext.html>



HOW TO CHOOSE A TECHNOLOGY (2021)

"How to choose a technology" is a performance that tries to critique the prescriptive nature of design methods practiced in the industry, especially to predict the future.

A ballot box is kept for three days at the event where visitors write their favourite future technologies. At the end of three days, the ballot is opened and the technologies are presented to the audience. These post-it notes (typical design thinking) are then pasted on a dart-board. Audience is now asked to aim. The post-it with most darts is chosen and discussed.





A solar car, which is a flat, rectangular vehicle covered in solar panels, is driving on a two-lane asphalt road. A person is visible sitting on the car. In the distance, another similar solar car is visible on the road. The road is flanked by dry grass and trees under a clear sky.

HOW TO BUILD A SOLAR CAR AND CROSS AUSTRALIA WITH IT (2011)

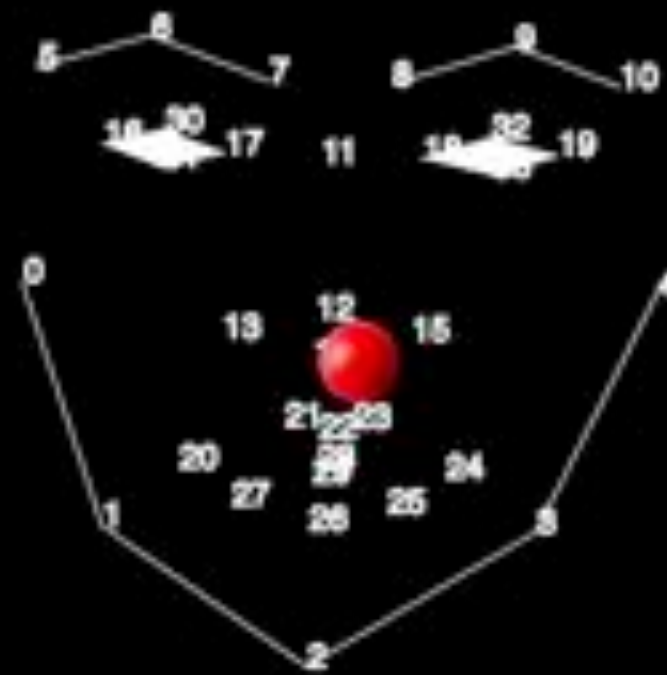
World Solar Challenge, Australia, 2011

3000 km race from Darwin — Adelaide only on a solar powered vehicle

<http://rohitg.in/2014/04/01/Solaris/>

INNOCENT INTERPRETATIONS SKETCHES (2019)

This series of work critiques the nature of emotional AI and the conversion of emotions to numbers. This was used as a stimulus to tell stories to encourage designers to critically understand emotion technologies and design interfaces for them.



<http://rohitg.in/portfolio/works/innocent.html>

DAILY ANIMALS (2020)

Media Lab de Prado, Madrid

Daily animals is a realtime noise visualisation that changes based on the time of the day. It tries to critique the rituals daily routine forces upon us as a shared system and how that affects the city. The noise amplitudes were programmed by logging the traffic on the roads. This helped in telling stories around ritualisation within the city.



WARLI POSE (2021)

Warli Painting is an ancient tribal art form that shows tribal cultural practices. I was enamoured by the simplicity and directness of the communication. It represents a communal social scene. In my view, this connects the dying past to the exploding present.

As a next phase, I will try to implement modern cultural scenes using Warli art as a way to critique the evolving social systems that used to protect an individual. What if Warli Painting were to be drawn today?

Uses : P5LIVE, p5js, and ml5js



COMPUTATIONAL MEDITATIONS (2021)

Computational Meditations is a performance where the artist guides the audience through a meditation routine using music that is computed in real-time by the artist. Certain elements of the meditation like breathing tempo and sonic tension are influenced by the generated audio.

The performance tries to create a relaxed feeling in the audience through an audio experiment and critiques the modern mindfulness movement. The performance is followed by a brief discussion on experience and ethics of mindfulness.

Uses : SuperCollider and Tidal Cycles





THANK YOU

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